Low-Power, Low-Cost CMOS Direct-Conversion Receiver Front-End for Multistandard Applications

Jusung Kim, *Member, IEEE*, and Jose Silva-Martinez, *Fellow, IEEE*

***Abstract*—A broadband CMOS direct-conversion receiver with on-chip frequency divider has been integrated in a 0.13-μm CMOS process. The key feature of the proposed receiver front-end is a single low-noise transconductance amplifier (LNTA) driving a current-mode passive mixer terminated by a low-input-impedance transimpedance amplifier (TIA). The receiver chain has improved robustness to out-of-band interference and outstanding linearity. We employ a broadband common-gate (CG) LNTA with dual feedback to improve both gain and noise figure (NF) without breaking the fixed relationship between input impedance, transconductance gain, and load impedance. A LNTA load impedance boosting technique suppresses noise-amplification due to TIA, commonly found in passive mixers. The core circuit (RF and baseband signal path) consumes only 13 mW, and the prototype receiver achieves $>$22.4-dB conversion gain, $<$8.3 dB NF, and ≥ -1.5 dBm IIP$_3$ from 1.4 to 5.2 GHz. Maximum conversion gain of 24.3 dB and minimum NF of 6.5 dB are achieved at 1.4 and 2 GHz, respectively. The chip active area is 1.1 mm^2 with the entire RF signal path operated from a 1.2-V supply. The LO portion is biased from a 1.5-V supply.**

***Index Terms*—Broadband receiver, CMOS, common-gate amplifier, direct-conversion, feedback amplifier, low-noise amplifier (LNA), low-nose transconductance amplifier (LNTA), operational transconductance amplifier (OTA), passive mixer, transimpedance amplifier (TIA), wideband receiver.**

I. Introduction

STRONG demand for multistandard receivers encouraged the scientific community to find architectural solutions to maximize the SOC integration and reduce the bill of material (BOM). In particular, direct-conversion receivers have been widely adopted in state-of-the-art wireless communication systems for their higher level of integration and baseband reconfigurability. Nonetheless, its critical drawbacks such as $1/f$ noise, dc-offset, and high requirement for matching still prove challenging and mandate both system- and circuit-level remedies.

The CMOS technology trend towards aggressive scaling reduces area and power consumption with sufficiently high f_t to accommodate most of the existing commercial applications under 10 GHz. Compatibility with the digital part of the transceiver mandates the use of advanced (scaled down) CMOS process. However, deep-submicron CMOS transistors generate more flicker noise ($1/f$ noise) than conventional CMOS process and the only way to mitigate the low-frequency noise from device perspective is to increase the device size, but this approach does not take full advantage of scaled CMOS process properties. The most promising way to alleviate the $1/f$noise issue is to have zero dc current in the switching core [1], [2]. Although it has already been shown that nonzero time-varying current with zero average (dc) value can still generate flicker noise [3], the passive mixer shows orders of magnitude lower flicker noise compared to the Gilbert-cell mixer which requires dc current.

The receiver architecture based on a single low-noise transconductance amplifier (LNTA) driving a current-mode passive mixer loaded by a low-impedance transimpedance closed-loop amplifier has been widely adopted recently due to its beneficial feature in terms of noise and linearity [2], [4]–[6]. The architecture shows a drastic reduction in flicker noise due to the current-mode passive mixer devoid of static current. Further optimization can be done on the TIA after down-conversion by increasing its device size for flicker noise reduction and burning more current for thermal noise reduction. Regarding linearity, the distortion associated with a large voltage swing present in Gilbert-cell mixer and voltage-mode passive mixer is minimized in this architecture thanks to the use of TIA's virtual ground. Out-of-band interference (OBI) performance is improved by avoiding large voltage swing before the signal experiences first-order low-pass filtering via the I–V conversion on TIA.

However, the design of a broadband receiver based on the above architecture remains challenging in several respects. Since there is a single V–I conversion before reaching the mixer, the LNTA's transconductance should be sufficiently large to guarantee a low system noise-figure (NF). Inductor-degenerated LNTA can be adopted in narrow-band implementations [4], [6], [7], where large transconductance ($>$20 mS) with 50-Ω input impedance synthesis is allowed. On the other hand, a broadband LNTA (e.g., shunt-feedback LNTA and conventional CG-LNTA) has a fixed relationship between input impedance, transconductance gain, and load impedance. Thus, LNTA transconductance is limited by the input-matching condition. Low mixer input impedance creates another difficulty because feedback-based amplifiers rely on large loop gain that usually the LNTA cannot provide. In addition, noise amplification due to TIA [3] can be significant in broadband design, whereas LC tank load for LNTA is utilized for narrow-band design to mitigate the switch capacitor effects [4], [6],

Manuscript received April 03, 2012; revised April 20, 2013; accepted May 16, 2013. Date of publication June 19, 2013; date of current version August 21, 2013. This paper was approved by Associate Editor Venu Gopinathan.

J. Kim was with the Analog and Mixed Signal Center, Texas A&M University, College Station, TX 77843 USA. He is now with Qualcomm Technologies Inc., San Diego, CA 92121 USA (e-mail: jusungk@qti.qualcomm.com).

J. Silva-Martinez is with the Analog and Mixed Signal Center, Texas A&M University, College Station, TX 77843 USA (e-mail: jsilva@ece.tamu.edu).

Color versions of one or more of the figures in this paper are available online at http://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/JSSC.2013.2265781

0018-9200 © 2013 IEEE

Authorized licensed use limited to: ANNA UNIVERSITY. Downloaded on May 22,2025 at 05:38:35 UTC from IEEE Xplore. Restrictions apply.

[8]. Frequency-dependent input impedance of TIA deserves a careful attention as well, especially when used in wideband systems. The feedback gain drops at higher frequencies with the increase in TIA input impedance. The conventional solution to the high-frequency input impedance degradation is to add a shunt capacitor to the TIA input node [6], [8]. However, this approach: 1) drastically reduces the open-loop OTA high frequency gain; 2) causes TIA noise peaking; and 3) degrades filtering of out-of-band interferences [2].

In this paper, we present a broadband, low-power, low-cost receiver front-end with a dual feedback CG-LNTA driving a current-mode passive mixer loaded by a virtual-ground generated by a wideband feed-forward-compensated TIA. The CG-LNTA utilizes capacitor cross-coupling and positive feedback from the cascode node of the amplifier such that the LNTA overcomes the design tradeoff among input impedance, transconductance gain, and LNTA load impedance. The quadrature sampling mixer topology reported in [9] is used to realize the 25% duty-cycle signals with a 50% quadrature LO. The quadrature mixing scheme significantly saves power consumption and improves noise performance. For the TIA implementation, a feed-forward-compensated broadband OTA design maintains wideband low input impedance relative to the Miller-compensated OTA with pole separation. The feed-forward OTA further benefits from better out-of-band filtering. Some of the standards that the proposed receiver front-end can support are the wireless network (e.g., 802.11b and g at 2.4-GHz ISM band, and 802.11a at 5.2 GHz) and the short-range communication standard (e.g., Bluetooth at 2.4-GHz ISM band). NF requirements on wireless network (Wi-Fi) are 14.8 dB for 802.11b and 7.5 dB for 802.11a and g [10]. Linearity of the proposed work has sufficient margin to accommodate those standards.

This paper is organized as follows. The system description is discussed in Section II. Section III describes the chip design details for CG-LNTA, quadrature sampled mixer, feed-forward OTA, and frequency divider, followed by measurement results in Section IV. Concluding remarks are given in Section V.

II. System Description

A. Receiver Architecture

The receiver block diagram is depicted in Fig. 1. The receiver front-end employs a direct-conversion architecture because of its lower cost and power as well as its baseband reconfigurability. The receiver front-end comprises an LNTA, two separate passive mixers for in-phase (I) and quadrature-phase (Q), and a transimpedance single-pole filter. The local oscillator (LO) generator consists of a divide-by-two circuit to generate a 50% duty-cycle LO from the external $2f_{\rm LO}$ source, and chains of self-biased CMOS inverters to provide rail-to-rail swing to the passive mixer. Since the LNTA output is current (transconductance), 50% duty-cycle LO with a relative delay of 1/4 period gives 6-dB gain loss and significant NF increase due to the overlap between LO-I and LO-Q pulses. The 25% duty-cycle LO scheme [11] provides isolation in time between I- and Q- current-path and, therefore, prevents conversion gain reduction and subsequent NF increase. However, the

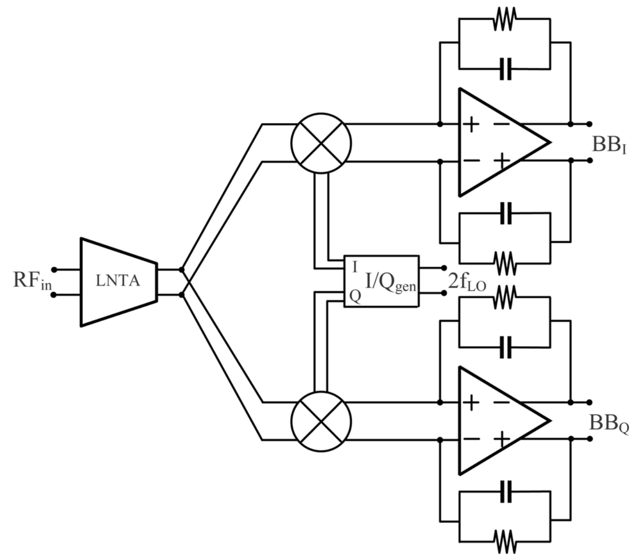


Fig. 1. Receiver architecture: LNTA, passive mixer, transimpedance single-pole filter with I/Q generator (divide-by-two and mixer buffer).

power dissipation to generate the required 25% duty-cycle LO might consume prohibitively large power. ANDing operation to provide 25% duty-cycle LO signals can be generated using the mixer switches instead of the LO [9]. Specifically, a cascade of mixer switches implements a nonoverlapping switching mixer between the I and Q paths with a 50% duty-cycle LO. The increase in on-resistance due to quadrature sampling does not impact the receiver performance significantly because we employ a LNTA load impedance boosting technique with wideband low-input-impedance TIA. All of the RF and LO paths are differential to enhance receiver robustness to power supply and substrate noise. Furthermore, the use of a current-mode passive mixer and careful design of baseband TIA significantly mitigates the flicker noise in the proposed architecture. The low input impedance presented by the TIA virtual ground reduces the distortion of the passive mixer [12]. The TIA's low-pass filtering mitigates the impact of OBI, which is more important for broadband receivers than narrowband ones with RF band-selection [5].

The most critical parameter of the receiver front-end is its sensitivity performance. To ensure low NF, large amplification with low-noise performance early in the chain is mandatory [13]. Thus, the RF transconductance of the receiver front-end should be large, and this requirement has been met in previous works [4], [6], [7] employing inductor-degenerated LNAs. Note that the effective transconductance (G_m) of the inductor-degenerated LNAs is independent of its input impedance as

$$G_m = Q_{\rm in} \cdot g_m = \frac{\omega_T}{2 \cdot \omega_o \cdot R_S} \tag{1}$$

where R_S is the source impedance and ω_o and ω_T are the operating and unity current gain frequencies, respectively. However, broadband LNAs (e.g., shunt-feedback and common-gate LNTA) present a fixed relationship between input impedance, transconductance gain, and load impedance. A shunt-feedback LNTA is not suitable for this application for the following reasons. Firstly, it is difficult to achieve sufficiently low NF and

Authorized licensed use limited to: ANNA UNIVERSITY. Downloaded on May 22,2025 at 05:38:35 UTC from IEEE Xplore. Restrictions apply.

the required input impedance with a reasonable current consumption. For instance, the work in [14] shows excellent broadband performance but consumes 42 mW. In addition, shunt-feedback LNTAs require large loop gain but, due to the low-input-impedance of passive mixer and TIA, LNTA's loop gain is drastically reduced when cascaded with the mixer.

Despite CG-LNTA's desirable broadband properties [15], high NF under input matching condition prevents its extensive use. Under input-matching conditions, its effective transconductance is expressed as [16], [17]

$$G_m = \frac{1}{2}g_m = \frac{1}{2R_S}. \tag{2}$$

In a 50-Ω R_S system, this requirement dictates that $G_{m,\text{CG-LNTA}}$ is only 10 mS, making it unsuitable for the proposed architecture where larger transconductance is needed to ensure sufficient system NF performance. In the proposed architecture, dual feedback, shunt-series negative feedback [17], and shunt-shunt positive feedback [18] are utilized to decouple the tradeoff between critical design parameters and to enable higher effective transconductance (>20 mS). Woo *et al.* [19] reported similar architecture with LNA only but it is not applicable to the proposed system with low mixer input impedance.

The proposed architecture features efficient current transfer between the LNTA and passive mixer, and improved linearity by employing an LNTA load-impedance-boosting technique and a wideband low-input-impedance OTA. Given the limited power supply of 1.2 V, a shunt-peaking inductor boosts the load impedance at the receiver bandwidth. An expression for finding the optimum inductor size is derived in Section III-A. Low TIA input impedance over a wideband is ensured with a feed-forward compensation technique, where the dominant pole need not be placed at low frequency as in a Miller-compensated OTA. Large swing due to OBI is significantly reduced resulting in better large-signal linearity (e.g., $P_{1\,\text{dB}}$) performance.

B. Receiver Performance

Fig. 2 shows the RF transconductor used for the receiver performance analysis. The RF transconductor is based on a dual-feedback LNTA and is cascaded with the quadrature-sampling mixer switches. The current transfer from LNTA to the quadrature-sampling switch is enhanced with a load boosting inductor. After switching, frequency translated signal flows into a single-pole *RC* low-pass network. The conversion gain of the proposed architecture can then be expressed as

$$\frac{V_{\text{out}}(f_{\text{out}})}{V_{\text{in}}(f_{\text{in}})} \approx \frac{\sqrt{2}}{\pi} \cdot \left|\frac{Z_{\text{in}}}{Z_{\text{in}}+R_S}\right| \cdot g_{mn} \cdot (1+A_n) \cdot |Z_F| \tag{3}$$

where f_{out} is the output frequency after down-conversion, f_{in} is the RF input frequency, A_n is the negative feedback factor ($A_n = (C_C - C_{\text{gs}})/(C_C + C_{\text{gs}}) \approx 1$), and $\sqrt{2}/\pi$ (= c) is the conversion gain of the 25% switching mixer. Z_{in} is the input impedance of LNTA, and Z_F is the TIA feedback impedance

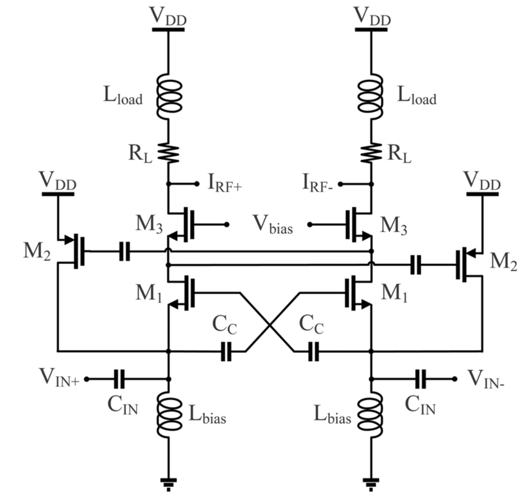


Fig. 2. Schematic of dual-feedback CG-LNTA (LNTA block in Fig. 1).

expressed as

$$Z_F = \left(\frac{R_F}{1 + j2\pi f_{\text{out}} R_F C_F}\right) \tag{4}$$

where R_F and C_F are feedback resistance and capacitance of the TIA shown in Figs. 1 and 10. Under perfect impedance match, effective transconductance (G_m) becomes equal to $(g_{mn}/2) \cdot (1 + A_n)$, which can be made larger than $1/2R_S$ while maintaining the input impedance match.

The dominant noise sources are M_1 (g_{mn}), M_2 (g_{mp}), M_3 (g_{m3}), R_L, R_F, and V_{amp} (input-referred noise of OTA). Noise of the cascode device M_3 is taken into account since its noise is injected into the circuit through M_2. Noise contributions due to the switches are neglected because of low average on-resistance and small dc current [20]. The overall noise at the output of the TIA due to these dominant noise sources can then be obtained as

$$V_{R_S}^2 = n \times 4kTR_S \left(\frac{g_{mn}(1+A_n)}{H}\right)^2 c^2 |Z_F|^2 \tag{5}$$

$$V_{M_1}^2 = n \times 4kT\frac{\gamma}{\alpha} g_{mn} \left(\frac{1}{H}\right)^2 c^2 |Z_F|^2 \tag{6}$$

$$\begin{aligned} V_{M_2+M_3}^2 &= n \times 4kT\frac{\gamma}{\alpha} g_{mp}(1+A_p) \\ &\quad \times \left(\frac{g_{mn}(1+A_n)R_S}{H}\right)^2 c^2 |Z_F|^2 \end{aligned} \tag{7}$$

$$V_{R_L}^2 = n \times 4kTR_L \left(\frac{1}{R_L + j2\pi f_{\text{in}} L}\right)^2 c^2 |Z_F|^2 \tag{8}$$

$$V_{R_f}^2 = 4kTR_F \left|\frac{1}{1 + j2\pi f_{\text{out}} R_F C_F}\right|^2 \tag{9}$$

$$V_{\text{amp}}^2 = v_{n,amp}^2 \left|1 + \frac{2Z_F}{Z_{\text{LNTA}}}\right|^2 \tag{10}$$

where γ and α are bias-dependent parameters [21], and n (noise aliasing factor) and H are

$$n = 2\left(1 + \frac{1}{3^2} + \frac{1}{5^2} + \cdots\right) = \frac{\pi^2}{4} \tag{11}$$

$$H = 1 + g_{mn}(1+A_n)(1-A_p)R_S \tag{12}$$

Authorized licensed use limited to: ANNA UNIVERSITY. Downloaded on May 22,2025 at 05:38:35 UTC from IEEE Xplore. Restrictions apply.

where $A_p = g_{mp}/g_{m3}$ is the positive feedback factor. Thus, the double-sideband (DSB) NF is approximately computed as

$$\mathrm{NF_{DSB}} = \frac{1}{2}\left(\frac{V_{R_S}^2 + 2\left(V_{M_1}^2 + V_{M_2+M_3}^2 + V_{R_L}^2 + V_{R_f}^2\right) + V_{\mathrm{amp}}^2}{4kTR_S\left(\frac{\mathrm{V_{out}}(f_{\mathrm{out}})}{\mathrm{V_{in}}(f_{\mathrm{in}})}\right)^2}\right) \tag{13}$$

$$= \frac{1}{2}(n + F_{M_1} + F_{M_2+M_3} + F_{R_L} + F_{R_f} + F_{\mathrm{TIA}}) \tag{14}$$

$$\approx \frac{n}{2} + n\frac{\gamma}{\alpha}\frac{1-A_p}{1+A_n} + n\frac{\gamma}{\alpha}g_{mp}(1+A_p)R_S + n\frac{4R_S}{R_L}(1-A_p)^2\left|\frac{R_L}{R_L + j2\pi f_{\mathrm{in}}L_{\mathrm{load}}}\right|^2 + \frac{4R_S}{R_F}(1-A_p)^2\frac{1}{c^2} + \frac{2R_S}{kT}v_{n,\mathrm{amp}}^2(1-A_p)^2\frac{1}{c^2|Z_{\mathrm{LNTA}}|^2}. \tag{15}$$

The factor of 2 in the numerator in (13) is due to the double-balanced mixer operation, and $Z_F \gg Z_{\mathrm{LNTA}}$ is used for the approximation in (15). TIA noise amplification due to the switched-capacitor (SC) effects is more pronounced at higher RF input frequency. The value of this SC impedance with L_{LNTA} neglected is [2], [8]

$$R_{\mathrm{LNTA}} = \frac{1}{4f_{\mathrm{LO}}C_{\mathrm{par}}} \tag{16}$$

where the factor of 4 arises from the parallel connection of the SC resistors in the quadrature (I/Q) mixer. While a narrowband receiver can null this noise effect employing an *LC* tank load for LNTA [4], [6], [7], a broadband receiver suffers degraded NF.

The receiver NF improves as a result of both negative (A_n) and positive (A_p) feedback loops. A_p has more impact on NF than A_n because positive feedback improves LNTA signal-to-noise ratio (SNR) as well as its effective transconductance. Neither the conventional CG-LNTA nor the negative feedback (A_n) loop enhance the amplifier current gain, and, thus, effective transconductance of the system remains equal to $(1/2R_S)$ while satisfying matching condition, whereas the effective transconductance with the positive feedback (A_p) is proportional to g_{mn}. The benefits of A_p become evident from (15) because the noise contribution due to M_1, R_L, R_F, and V_{amp} are drastically suppressed when A_p approaches unity. However, noise contribution due to M_1 can only be minimized with larger negative feedback (A_n). Fig. 3 shows the double sideband (DSB) NF performance when different types of CG-LNTAs are used; these results support the aforementioned remarks. In this design, $A_p = 0.5$ is used for positive feedback gain, but the results can be extrapolated to any other A_p value. For system-level noise analysis, we assume a OTA flicker noise corner of 200 KHz and an *RC* filter pole at 1.92 MHZ (UMTS signal bandwidth). Fig. 4 shows the receiver tradeoff between NF and current consumption as a function of the positive feedback factor (A_p). High A_p improves the NF by enhancing the LNTA's SNR and RF transconductance (G_m) at the expenses of power consumption and potential instability. We also assume

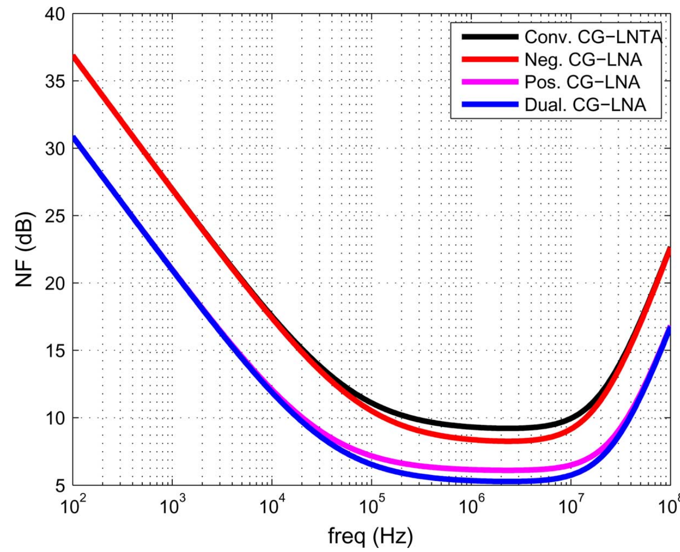


Fig. 3. Receiver NF performance with different broadband CG-LNTAs.

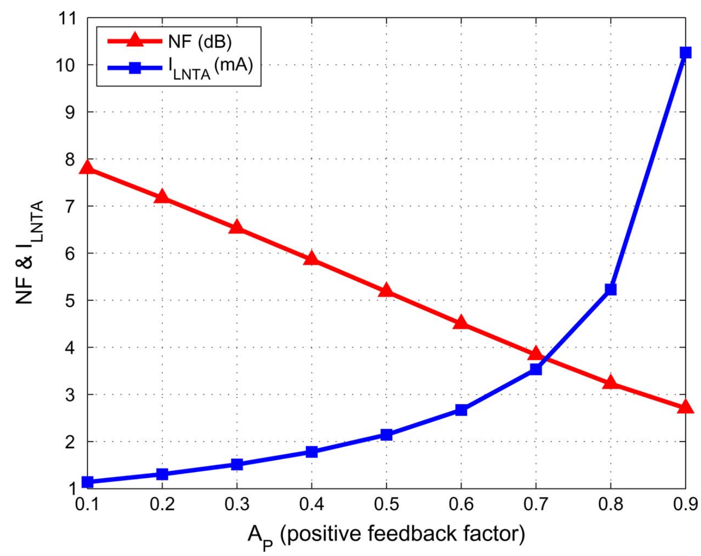


Fig. 4. NF and power consumption (I_{LNTA}) for different positive gain factors (A_p) with $v_{\mathrm{dsat}} = 0.2$ V.

the classic square-law model with overdrive voltage (v_{dsat}) of 0.2 V. Hence, LNTA's dc current can be expressed as

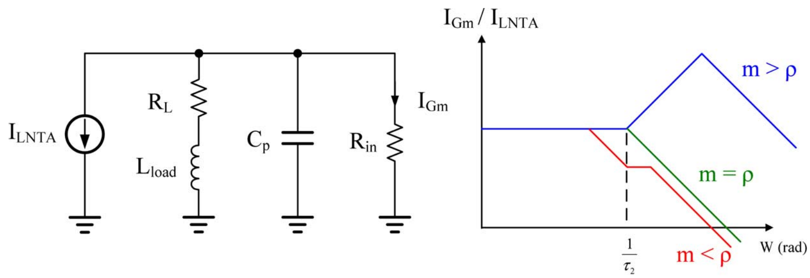
$$I_{\mathrm{LNTA}} = \frac{1}{2}(g_{mn} + g_{mp})v_{\mathrm{dsat}}. \tag{17}$$

III. Building Block Designs

A. Low-Noise Transconductance Amplifier

In Fig. 2, impedance-matching device M_1 provides the main forward signal path. g_m boosting is implemented with noiseless capacitor cross-coupling [17] with almost unity negative feedback gain. Shunt-shunt positive feedback through M_2 enables the LNTA current gain to be larger than 1 and at the same time increases the input impedance. Since the required positive feedback factor $(A_p = g_{mp}/g_{m3})$ is small, M_2 is biased with

Authorized licensed use limited to: ANNA UNIVERSITY. Downloaded on May 22,2025 at 05:38:35 UTC from IEEE Xplore. Restrictions apply.



(a) (b)

Fig. 5. (a) Equivalent circuit diagram for current-transfer optimization. (b) Transfer function versus relative location of two poles.

a large overdrive voltage $(V_{\rm dsat})$ then making its nonlinear effect negligible. The effective $V_{\rm dsat}$ of M_1 is reduced with the positive feedback [18] but it is not the dominant factor determining the receiver linearity performance. The previous works utilized positive feedback [18], [19] sensing the signal at the amplifier output and then fed back to the input. However, in the proposed architecture, the RF transconductor drives directly a low impedance mixer terminated by virtual ground and, therefore, feedback from the output cannot be employed. Instead, the signal is sensed at the cascode node (source of M_3), leading to a well-controlled loop gain determined by the ratio of two transconductances $(A_p = g_{mp}/g_{m3})$. Variability of A_p is small and can be expressed as

$$\frac{\Delta A_p}{A_p} = S^{A_p}_{g_{mp}} \frac{\Delta g_{mp}}{g_{mp}} + S^{A_p}_{g_{m3}} \frac{\Delta g_{m3}}{g_{m3}} = \frac{\Delta g_{mp}}{g_{mp}} - \frac{\Delta g_{m3}}{g_{m3}} \approx 0. \tag{18}$$

Constant-gm bias network can be utilized to further stabilize the LNTA operation over process and temperature variation although not implemented in this work. Direct coupling path from LNTA output to RF input is also avoided with the proposed architecture which improves the LO leakage. Possible increase of the parasitic capacitance at the input is nulled by the bias inductor $(L_{\rm bias})$.

The in-band LNA input impedance can be found as

$$Z_{\rm in} \approx \frac{1}{g_{mn}(1+A_n)(1-A_p)} \parallel sL_{\rm bias} \parallel \frac{1}{sC_{\rm par}} \tag{19}$$

where the parasitic capacitor $C_{\rm par}$ arises from the contributions of the input pad, M_1, M_2, and C_C.

A shunt-peaking inductor $(L_{\rm load})$ at the LNTA output enhances the current transfer into the mixer and minimizes the OTA noise contribution. Fig. 5(a) shows the equivalent circuit diagram used for broadband current-transfer optimization. $R_{\rm in}$ is the mixer input impedance when the switch is on and can be approximated as

$$R_{\rm in} = 2R_{\rm switch} + \frac{R_F}{1+A_{\rm OTA}} \tag{20}$$

where the factor of 2 is the result of the quadrature sampling mixer topology used for the proposed receiver, and $A_{\rm OTA}$ is the open-loop gain of the OTA. The effect of the parasitic capacitors at the OTA input is ignored to simplify the analysis. The current transfer into the mixer can be obtained as

$$\begin{aligned}\frac{I_{Gm}}{I_{\rm LNTA}} &= -\frac{\left(\frac{R_L}{R_L+R_{\rm in}}\right)\left(1+s\frac{L_{\rm load}}{R_L}\right)}{1+s\left(\frac{L_{\rm load}}{R_L+R_{\rm in}}+\frac{R_{\rm in}R_LC_P}{R_L+R_{\rm in}}\right)+s^2\frac{R_{\rm in}C_PL_{\rm load}}{R_L+R_{\rm in}}}\\ &= -\frac{\rho(1+s\tau_1)}{1+s\rho(\tau_1+\tau_2)+s^2\rho\tau_1\tau_2}\end{aligned} \tag{21}$$

where $\rho\ (= R_L/(R_L+R_{\rm in}))$ is low-frequency current-transfer gain, and $\tau_1\ (= L_{\rm load}/R_L)$ and $\tau_2\ (= R_{\rm in}C_P)$ are time constants present in the equivalent circuit. To facilitate subsequent derivations, we introduce a factor m $(= \tau_2/\tau_1)$. The damping factor (ζ) and the undamped natural frequency (ω_n) of the current transfer function can be derived from (21) as

$$\zeta = \frac{1}{2}\sqrt{\rho}\left(\sqrt{\frac{\tau_1}{\tau_2}}+\sqrt{\frac{\tau_2}{\tau_1}}\right) \geq \sqrt{\rho} \tag{22}$$

$$\omega_n = \frac{1}{\sqrt{\rho\tau_1\tau_2}}. \tag{23}$$

Depending on the relative location of the two poles, the transconductance follows the trends shown in Fig. 5(b). The value of $L_{\rm load}$ is upper limited to ensure $m > \rho$ as

$$L_{\rm load} < \frac{R_L \cdot R_{\rm in} \cdot C_P}{\rho}. \tag{24}$$

Notice that the bandwidth of the current transfer function without $L_{\rm load}$ $(= 1/(R_{\rm in} \parallel R_L)C_P)$ approaches $1/\tau_2$ when $R_L \gg R_{\rm in}$.

Since the proposed LNTA utilizes positive feedback, the stability issue needs to be carefully addressed. As a measure of the stability, Fig. 6 shows a simulated Rollett's stability factor (k-factor) [22] from low (1 kHz) to high (10 GHz) out-of-band frequencies. The minimum k-factors are 11.3 and 17.2 when the positive feedback path is enabled and disabled, respectively. The LNTA is unconditionally stable and conservative value of $A_p = 0.5$ in this design does not hurt the stability.

B. Switches

Fig. 7(a) shows the 25% duty-cycle mixer switches, which is drawn with single-ended RF input not to clutter the schematic. The double balanced topology presents the benefit of rejecting LO feed-through and noise due to the LO circuitry. Also, the double-balanced structure improves IIP2 performance with

Authorized licensed use limited to: ANNA UNIVERSITY. Downloaded on May 22,2025 at 05:38:35 UTC from IEEE Xplore. Restrictions apply.

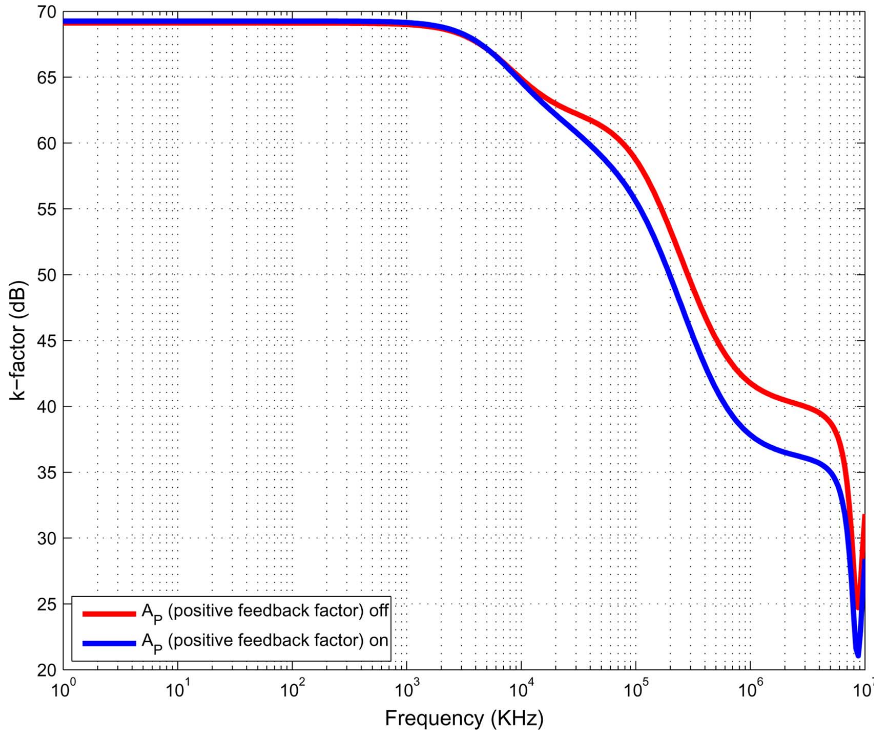


Fig. 6. Stability of the proposed LNTA (Rollett's stability factor in decibels).

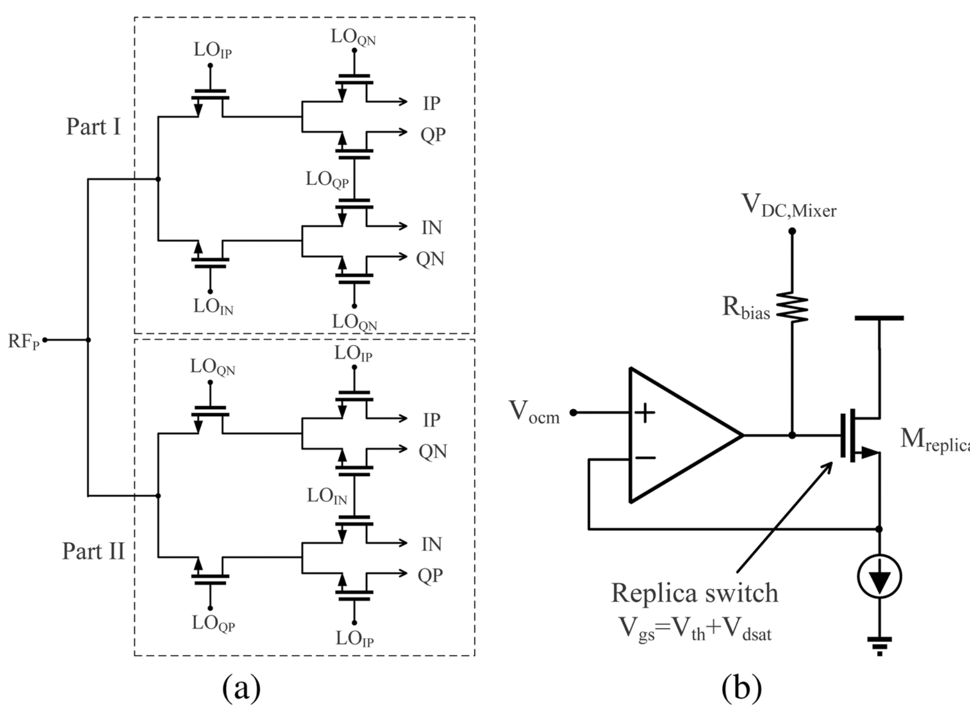


Fig. 7. (a) 25% duty-cycle quadrature sampling mixer (single-ended RF input used) [9]. (b) Replica circuitry for mixer switch dc bias.

less self-mixing effects [8], [23]. The switching topology in Fig. 7(a) realizes a 25% duty-cycle by ANDing two 50% I-Q LO signals with a relative delay of 1/4 period. Fig. 8 shows a timing diagram illustrating the AND operation of the quadrature switching topology. Switching parts I and II together achieve the equal loading from 50% I-Q LO, whereas each of them introduces I-Q mismatch, since in-phase LO drives one switch and the quadrature-phase LO drives two switches in switching part I and vice versa in switching part II. Potential residual side-band (RSB) performance reduction is prevented by this scheme. Previous works in [11] and [24] implement 25% duty-cycle switching with either 50% duty-cycle $2f_{\mathrm{LO}}$ or 25% duty-cycle f_{LO}, both of which require significant power consumed by the LO buffers.

Authorized licensed use limited to: ANNA UNIVERSITY. Downloaded on May 22,2025 at 05:38:35 UTC from IEEE Xplore. Restrictions apply.

Depending on the dc gate bias of the switches, the passive mixer can operate in either ON or OFF overlap of the mixer switches. ON overlap results in lowered conversion gain and, more importantly, noise amplification due to OTA becomes significant. It is evident from (10) that ON overlap of the switches results in a significant gain of OTA input-referred noise voltage (V_{amp}) due to the small on-resistance of the switches. On the other hand, OFF overlap results in linearity degradation [20], [25]. Fig. 7(b) shows the replica circuitry to bias the gate of the passive mixer switches. The replica transistor $(M_{\rm replica})$ is biased at weak inversion where its $V_{\rm gs}$ is near the threshold voltage $(V_{\rm th})$ and tracks with process variations. V_{ocm} represents the common-mode voltage used for the OTA common-mode feedback (CMFB) circuitry, which sets the dc bias voltage at the source (drain) of the passive mixer switches. The replica circuitry provides the dc gate bias for the switches at the threshold of conduction $(V_g \approx V_{\rm ocm} + V_{\rm th})$. The receiver simulation results show that the replica-bias circuit provides a DC gate voltage nearly equal to the optimal gate bias for minimum NF.

C. Baseband TIA

The TIA converts the down-converted current into voltage with very small input impedance, hence allowing the current-mode passive mixer to operate properly. To accommodate high-power blockers, the TIA includes a single-pole *RC* filter. The main limitation on the TIA design is to achieve high gain and wide bandwidth to maintain low input impedance without burning excessive power. The conventional solution [20], [26] employs two-stage OTA with Miller compensation and nulling resistor. This approach requires significant power consumption to achieve a large gain-bandwidth-product (GBW) and simultaneously maintain closed-loop stability. Furthermore, the inherent pole splitting property present in Miller compensation approach [27], [28] reduces the amplifier dominant pole significantly and so the TIA unity-gain frequency. As a result, the TIA input impedance shows inductive behavior after the amplifier's dominant pole and, thus, the signal swing due to jammer residing out-of-band increases. Previous works [6], [8], [29] added an extra capacitor at the virtual-ground node to suppress the inductive input impedance, but this scheme degrades the frequency dependent transconductance of the previous stage, reducing the gain and SNR of the receiver. Equation (10) manifests that the noise due to OTA $(V_{\rm amp})$ drastically increases with a capacitor present at TIA virtual-ground.

In the proposed receiver front-end, a feed-forward two-stage OTA [30] is employed to achieve the required TIA wideband low input impedance and ideal single-pole response. Fig. 9 shows the block diagram of the feed-forward compensation method along with the main design parameters. The feed-forward path due to g_{m3} introduces a LHP zero to compensate for the negative phase shifts from the poles generated by the two cascaded amplifiers used to generate large low-frequency gain. The amplifier transfer function based on Fig. 9(a) is obtained as

$$\frac{V_{\rm out}(s)}{V_{\rm in}(s)} = -\frac{(A_{v1}A_{v2}+A_{v3})\left(1+\frac{A_{v3}s}{(A_{v1}A_{v2}+A_{v3})\omega_{p1}}\right)}{\left(1+\frac{s}{\omega_{p1}}\right)\left(1+\frac{s}{\omega_{p2}}\right)}. \tag{25}$$

Miller compensation realizes stability at the expense of reduced dominant pole location and, therefore, the inductive behavior of TIA input impedance appears at rather lower frequency. Out-of-band peaking in the transimpedance transfer function moves to a lower frequency as well and degrades the *RC* filtering. Without effective *RC* filtering of the TIA, OBI would hamper the wideband receiver's linearity.

The TIA input-impedance and its transfer function can be derived from the schematic shown in Fig. 10; the equivalent circuit model includes the RC feedback components. After some simplifications, the TIA input impedance can be obtained as

$$\begin{aligned}\frac{v_{\rm in}}{i_{\rm in}} &= Z_{\rm in}(s)\\ &\approx \frac{R_F+R_2}{1+A_v}(1+sR_1C_1)\frac{A(s)}{D(s)}\end{aligned} \tag{26}$$

where A_v $(\approx g_{m1}R_1g_{m2}R_2)$ is the dc open-loop gain of the OTA, and $A(s)$ and $D(s)$ are

$$A(s) = \frac{1+s(R_2 \parallel R_F)(C_2+C_F)}{1+sR_FC_F} \tag{27}$$

$$D(s) = 1 + s\frac{g_{m3}}{g_{m1}g_{m2}}C_1 + s^2\frac{C_1C_2}{g_{m1}g_{m2}}. \tag{28}$$

The increase in TIA input impedance is due to the OTA dominant pole at $(1/R_1C_1)$ in contrast to $1/(R_1C_m(1+g_{m2}R_2))$ $(\ll 1/(R_1C_1))$ obtained with the Miller-compensated OTA $(C_m = \text{Miller capacitor})$. The damping factor (ζ) and the undamped natural frequency (ω_n) of $D(s)$ can be derived as

$$\zeta = \frac{1}{2}\kappa\frac{g_{m3}}{g_{m2}} \tag{29}$$

$$\omega_n = \sqrt{\frac{g_{m1}g_{m2}}{C_1C_2}} = \kappa\frac{g_{m1}}{C_1} \tag{30}$$

where $\kappa = \sqrt{(g_{m2}/C_2)/(g_{m1}/C_1)}$ is a proportionality constant that relates the gain–bandwidth product of each stage in the OTA. The characteristics of the TIA input impedance can then be adjusted by properly selecting the parameters of the frequency dependent term $D(s)$. Similar conclusions can be drawn for the TIA transimpedance gain expressed as

$$\frac{v_{\rm out}}{i_{\rm in}} \approx -\frac{A_v}{1+A_v}\frac{R_F}{1+sR_FC_F}\frac{G(s)}{D(s)} \tag{31}$$

where $D(s)$ is given in (28) and $G(s)$ is expressed as

$$G(s) = 1 + s\frac{1}{g_{m1}g_{m2}}\left(g_{m3}C_1 - \frac{C_F}{R_1}\right) - s^2\frac{C_1C_F}{g_{m1}g_{m2}}. \tag{32}$$

According to (32), an LHP zero due to g_{m3} is generated. Also, an RHP zero present in all integrators due to C_F is observed; if required, a series resistor can be used to move this zero to the left side of the S-plane. The schematic of the OTA, depicted in Fig. 11, employs a two-stage amplifier with g_{m3} used for feed-forward compensation. PMOS instead of NMOS differential pairs are chosen for reduced flicker noise performance and large input devices biased at moderately high current help reduce the equivalent input noise voltage [26]. Due to the active (g_{m3}) compensation, power and noise are of major con-

Authorized licensed use limited to: ANNA UNIVERSITY. Downloaded on May 22,2025 at 05:38:35 UTC from IEEE Xplore. Restrictions apply.

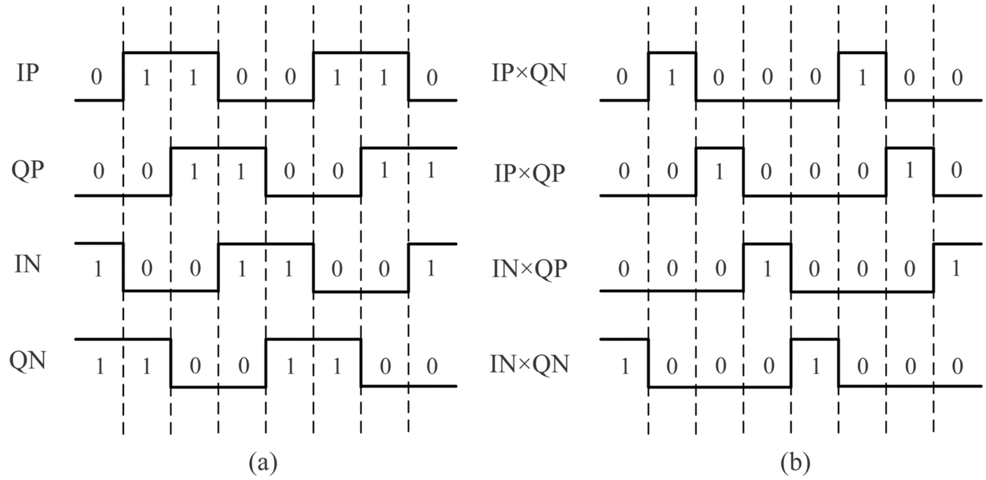
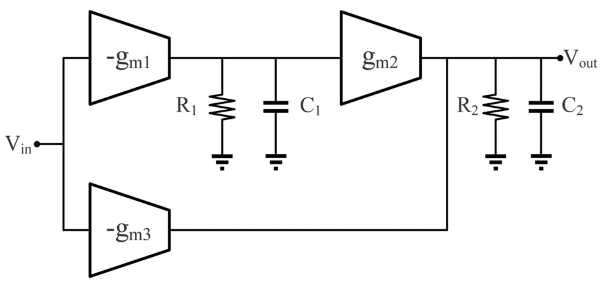


Fig. 8. (a) 50% duty-cycle LO. (b) 25% duty-cycle LO by ANDing of two 50% I-Q LO signals with a relative delay of 1/4 period.



(a)

$A_{v1} = g_{m1}R_1$	$\omega_{p1} = 1/R_1C_1$
$A_{v2} = g_{m2}R_2$	$\omega_{p2} = 1/R_2C_2$
$A_{v3} = g_{m3}R_2$	$\omega_{p3} = \omega_{p2}$

(b)

Fig. 9. (a) Block diagram of feed-forward compensation scheme for two-stage amplifier. (b) Design parameters.

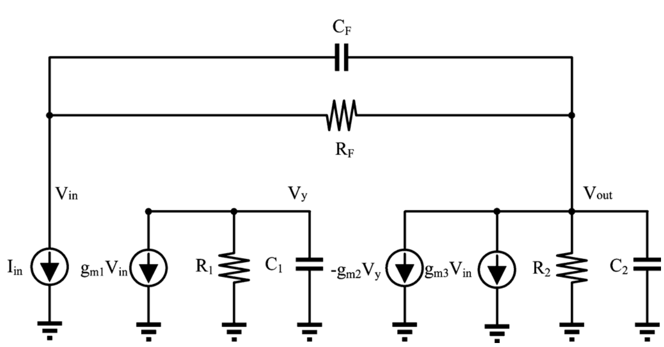


Fig. 10. Equivalent circuit model of TIA with feed-forward compensation.

cern. Assuming $A_{v1}A_{v2} \gg A_{v3}$, the input referred noise of the feed-forward OTA is given by

$$V_{amp}^2 = 8kT\frac{\gamma}{\alpha}\left(\frac{1}{g_{m1}} + \frac{g_{mn}}{g_{m1}^2} + \frac{g_{m2}+g_{m3}+g_{mp}}{(A_{v1}g_{m2})^2}\right) \quad (33)$$

Note that the noise due to g_{m3} is attenuated by A_{v1} within the bandwidth of the amplifier's first stage and its impact on

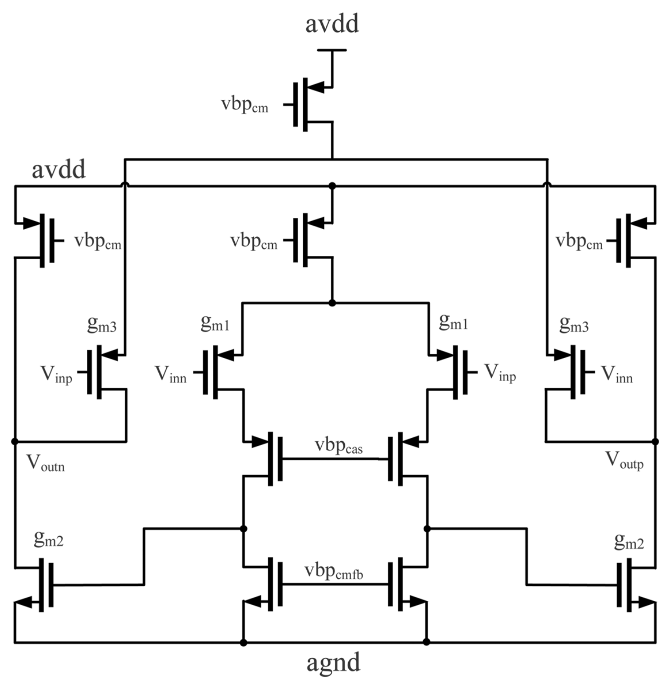


Fig. 11. Schematic of feed-forward compensated OTA (CMFB is not shown).

overall amplifier's noise is minimal. g_{m3} is set equal to g_{m1} in this design to obtain a Butterworth response for $D(s)$. Power efficiency is not degraded since the current required for the second-stage transconductance (g_{m2}) is re-used for active (g_{m3}) compensation.

The circuit shown in Fig. 11 was designed and simulated in Spectre RF. We design two different cases for the feed-forward OTA topology: 1) Butterworth response for $D(s)$ and 2) placing the frequency of the LHP zero such that it cancels the nondominant pole $(1/R_2C_2)$. Likewise, we consider two cases for the Miller-compensated OTA by placing the zero $(\omega_z = -(1)/(C_m(1/g_{m2} - R_Z)))$ at the following locations: 3) non-dominant pole $(\omega_{p2} \approx (g_{m2})/(C_1 + C_2))$ and 4) infinite frequency (∞). The TIA input-impedance for both amplifiers is plotted in Fig. 12. For the case of the proposed TIA, the inductive input impedance peaking is significantly reduced

Authorized licensed use limited to: ANNA UNIVERSITY. Downloaded on May 22,2025 at 05:38:35 UTC from IEEE Xplore. Restrictions apply.

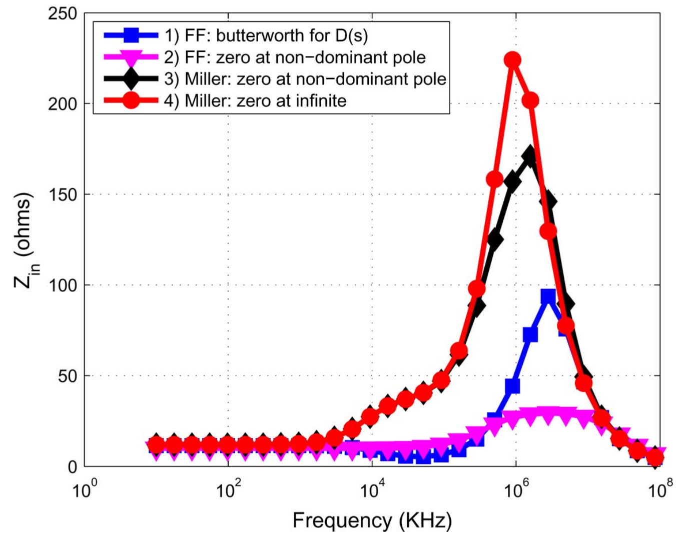


Fig. 12. TIA input impedance ($Z_{\rm in}$).

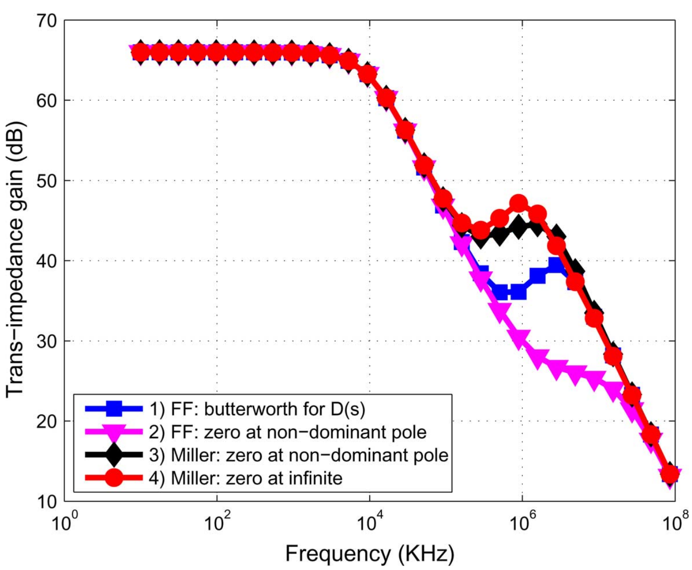


Fig. 13. TIA transimpedance gain $((v_{\rm out}(s))/(i_{\rm in}(s)))$.

and pushed to higher frequency range and enables broadband low input impedance. The Miller-compensated OTA can not maintain the low impedance behavior. Better OBI filtering is achieved with feed-forward OTA, as shown in Fig. 13.

D. Frequency Divider and LO Buffer

The LO clock generator is shown in Fig. 14. An off-chip $2f_{\rm LO}$ signal is injected through a balun. The divide-by-two circuitry generates 50% duty-cycle $f_{\rm LO}$ I- and Q- signals on-chip. Additional benefit is the reduced LO leakage from LO to RF ports. An edge-triggered master/slave flip-flop (DFF) implemented with current-mode logic (CML) implements the divide-by-two circuitry. The required operating frequency is between 2 and 14 GHz, twice the frequency of the internal LO frequency. The frequency divider is directly connected to the pin without any input buffer to minimize the dynamic power of the LO chain. Required clock swing for proper generation of $f_{\rm LO}$ on-chip is getting smaller until self-oscillating frequency

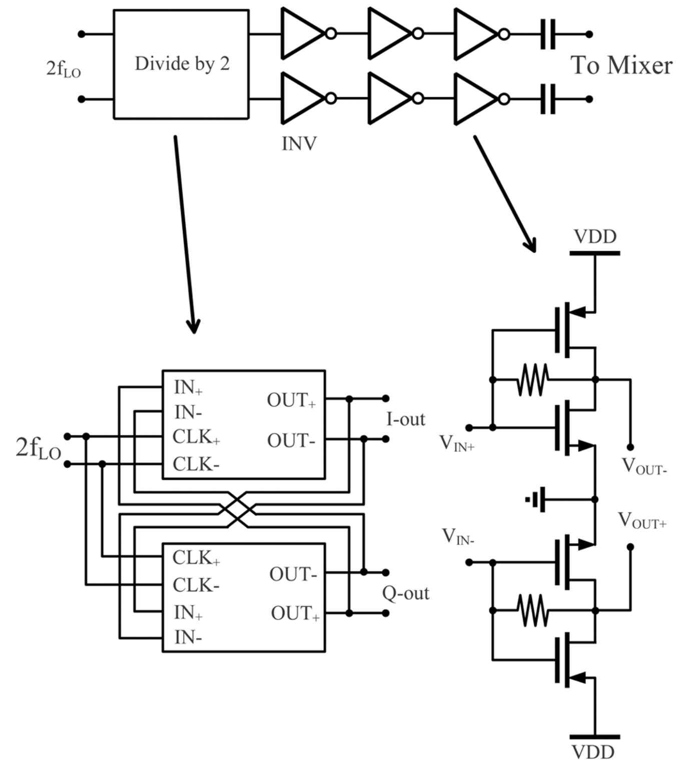


Fig. 14. LO generation circuitry (divider-by-two and chains of CMOS inverters).

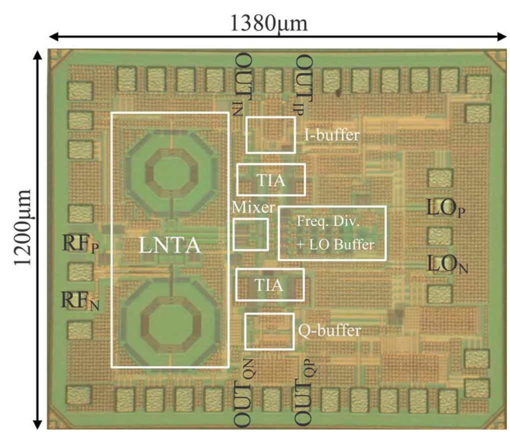


Fig. 15. Die photograph of the proposed receiver.

of divider based on CML latch, which obviates the need for input-buffer [31]. Chains of self-biased CMOS inverters provide rail-to-rail signals to improve passive mixer NF and IIP_2 performance [1], [23]. CML type inverters provide faster operating frequency than CMOS inverters. However, we chose CMOS inverters instead to achieve larger swing and to consume less power.

IV. Measurement Results

The circuit was designed in TSMC 0.13-μm CMOS technology and encapsulated in a quad flat no leads (QFN) package for connecting the dc pads and output. The RF and LO signals are applied using on-wafer probing to reduce the losses and mismatches introduced by the measurement setup. Fig. 15 shows the die photograph of the prototype receiver. The active

Authorized licensed use limited to: ANNA UNIVERSITY. Downloaded on May 22,2025 at 05:38:35 UTC from IEEE Xplore. Restrictions apply.

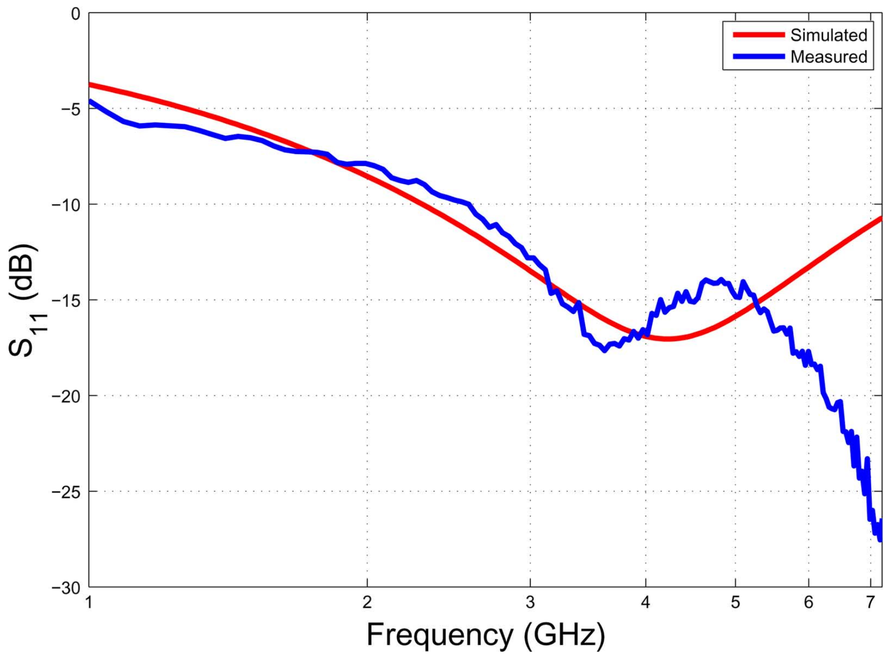


Fig. 16. Measured S_{11} of the proposed receiver.

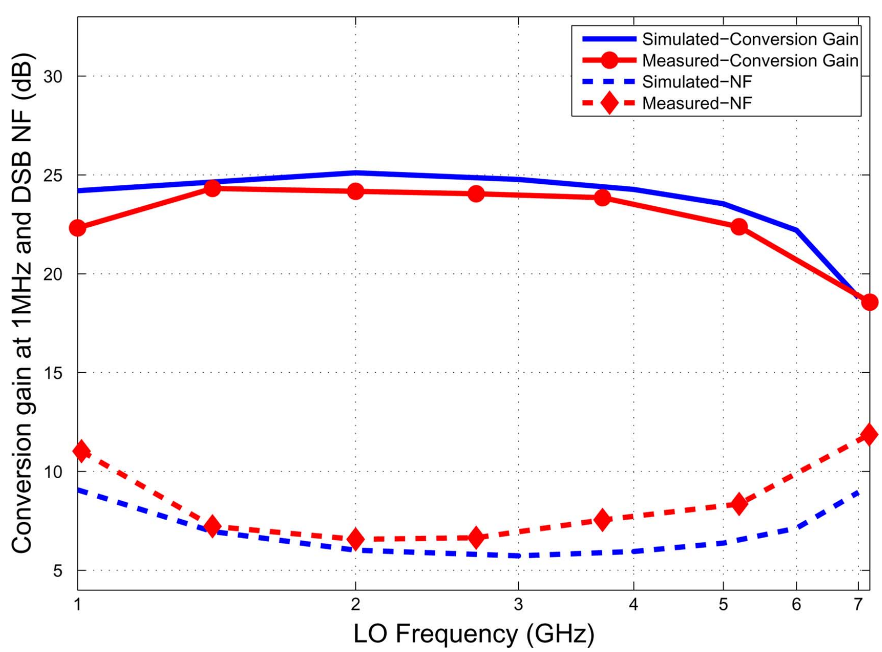


Fig. 17. Measured conversion gain $(A_{V,RX})$ at 1-MHz down-converted frequency and DSB NF of the proposed receiver.

area of the chip is 1.1 mm × 1 mm, and the CG-LNTA occupies 0.45 mm × 0.9 mm due to the differential inductors used for input matching and load boosting.

Fig. 16 shows both measured and simulated impedance matching performances (S_{11}). S_{11} of the prototype receiver was measured with RF signal wire-bonded since no on-wafer differential probe calibration was available. The simulated S_{11} is a parallel resonance $(L_{\text{bias}} \parallel C_{\text{par}})$ with low quality factor resulting in a wide bandwidth. The discrepancy between simulated and measured results shown at high frequency (>5 GHz) is due to the interaction between off-chip balun (Krytar's hybrid coupler, 4010180) at the RF input, the FR-4 board PCB, and package effects.

Fig. 17 shows the measured conversion gain $(A_{V,RX})$ at 1-MHz down-converted frequency and minimum in-band double-sideband (DSB) noise figure (NF). We measure a maximum conversion gain of 24.3 dB at 1.4-GHz LO frequency, and upper and lower 2-dB bandwidth is measured at 1 and 5.2 GHz, respectively. Both measured and simulated conversion gains are in good agreement; however, the measured conversion gain is about 1 dB lower than the simulated one across the frequency range. We attribute the discrepancy to: 1) imbalance of the off-chip balun at RF and LO inputs and 2) lower transconductance of the LNTA due to PVT variations. In fact, the measured dc-bias nodes show values between typical and slow corner for both NMOS and PMOS. The effect of

Authorized licensed use limited to: ANNA UNIVERSITY. Downloaded on May 22,2025 at 05:38:35 UTC from IEEE Xplore. Restrictions apply.

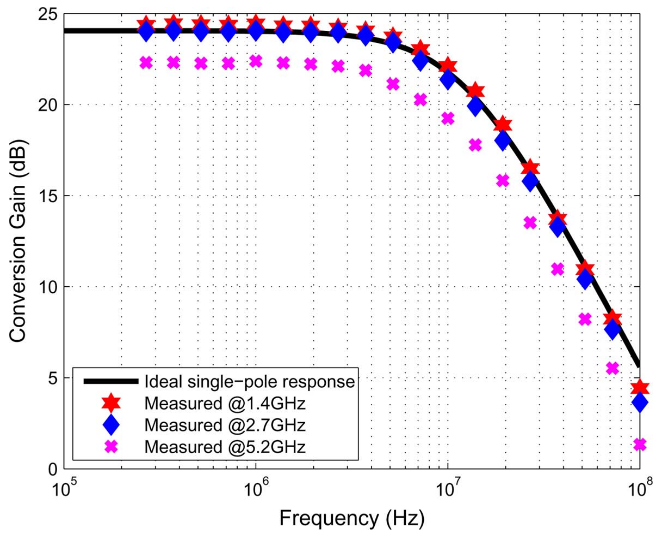


Fig. 18. Measured $A_{V,RX}$ versus down-converted frequency of the proposed receiver.

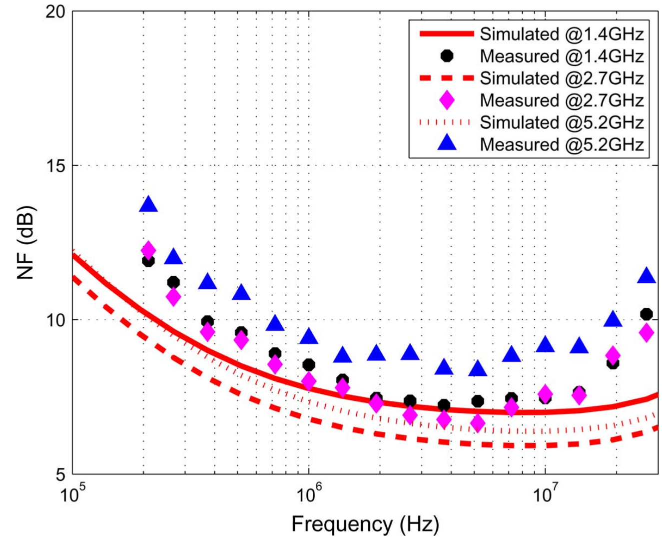


Fig. 19. Measured DSB NF versus down-converted frequency of the proposed receiver.

the output buffer and matching network on conversion gain is deembedded with the procedure detailed in Appendix A. The measured in-band DSB NF in Fig. 17 agrees well with the simulated results across the entire frequency range, and a minimum NF of 6.5 dB is measured at 2-GHz LO frequency. The NF discrepancy is mainly due to the lower gain of the LNTA due to the off-chip balun mismatch and lower transconductance, and the inaccuracies of the provided noise models. The degradation of $A_{V,RX}$ and NF after 5 GHz is noticeable but is not due to the bandwidth limitation of the receiver signal path. At higher LO frequencies, the LO buffer driving the passive mixer switch shows slow rising and falling times as well as reduced signal swing. The receiver bandwidth limitation arises from: 1) the frequency divider operating at twice the RF frequency and 2) LO buffer limitations.

Fig. 18 shows the measured $A_{V,RX}$ versus down-converted frequency. The measurements were performed at low- (1.4 GHz), mid- (2.7 GHz), and high- (5.2 GHz) frequency bands. The ideal single-pole response is included for reference. Due to the feed-forward-compensated OTA, almost ideal single-pole filtering is observed up to 100 MHz; as expected, out-of-band peaking is not observed. The DSB NF versus IF frequency was also measured, and the results are shown in Fig. 19. NF at low IF frequency degrades more rapidly than the simulated NF. This behavior is caused by an inaccurate flicker noise model and the noise contribution from the subsequent blocks; i.e., output buffer and measurement equipment. NF increment at high IF frequencies is expected due to the single pole TIA amplifier. The discrepancy between measured and simulated NF at these frequencies is attributed to a higher RC time constant used for single pole filtering in baseband.

Fig. 20 shows in-band IIP_3 measurement result at $f_{\mathrm{LO}} = 2.7$ GHz. Two tones with $P_{\mathrm{in}} = -25$ dBm at 3- and 5-MHz offset from the LO frequency are injected, and the in-band intermodulation is measured. Both the third-intermodulation tones at $2f_2 - f_1$ and $2f_1 - f_2$ are in-band and thus are not filtered in baseband. P_{in} of the two tones is swept and its fundamental

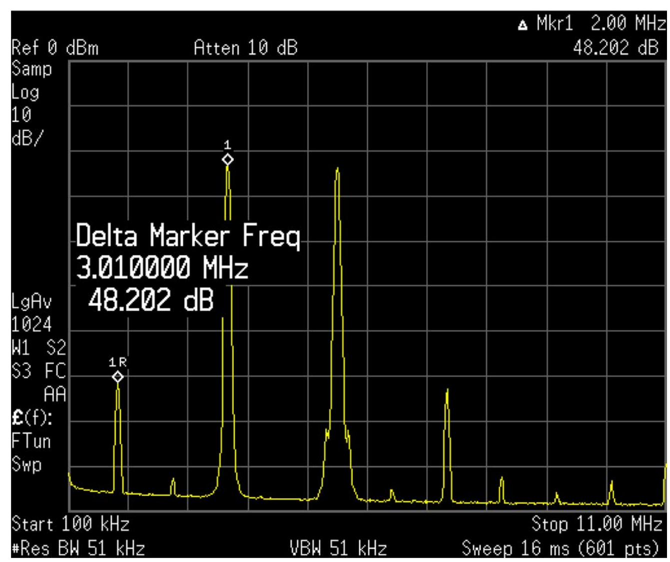


Fig. 20. Two-tone measurement at $f_{\mathrm{LO}} = 2.7$ GHz ($P_{\mathrm{in}} = -25$ dBm).

and third-intermodulation power are measured; the results are shown in Fig. 21. Higher order (fifth, seventh, and higher) intermodulation components appear when $P_{\mathrm{in}} > -20$ dBm. This large-signal nonlinear behavior can be improved with higher supply voltage and a class-AB output stage for the OTA.

Small-signal and large-signal linearity performance of the receiver is measured through IIP_3 and $P_{1\,\mathrm{dB}}$ as a function of the LO frequency. Due to the limited capability of two-tone generating measurement equipment, the linearity test was performed from 1 to 3 GHz. Fig. 22 shows that the minimum IIP_3 in the 1–3 GHz frequency range is -1.5 dB with variations of 2 dB. At low LO frequency, IIP_3 improves due to the impedance mismatch at the input as well as lower conversion gain. Both the in-band and out-band $P_{1\,\mathrm{dB}}$ shows similar trend to that of IIP_3 with respect to LO frequency. Out-band $P_{1\,\mathrm{dB}}$ is better than in-band $P_{1\,\mathrm{dB}}$ by 1 dB. However, the benefit of filtering on

Authorized licensed use limited to: ANNA UNIVERSITY. Downloaded on May 22,2025 at 05:38:35 UTC from IEEE Xplore. Restrictions apply.

TABLE I
COMPARISON TO RECENTLY PUBLISHED WORKS

	Technology	Frequency [GHz]	Gain [dB]	NF [dB]	IIP_3 [dBm]	Power [mW]	Supply [V]	LNTA Architecture	Area [mm^2]
[6]	0.13μm CMOS	5.15-5.825	26	3.5	-2	36* 36**	1.2* 2.5***	inductor-degenerated	1.8†
[4]	0.18μm CMOS	0.869-0.894	44	2.4	4.65◇	35.7*	2.1	inductor-degenerated	2.25
[2]	0.18μm CMOS	1.55-2.3	22.5-25	7.7-9.5	≥7	10*	2	CG	Not reported
[32]	90nm CMOS	0.1-3.85	20	8.4-11.5‡	-3.23◇◇	9.8*	1.2	CG	0.88
[20]	0.13μm CMOS	0.9-2.3	34.5-35.5	9.5-11.5	4-11	30-36 (total)	1.5	differential pair (resistive match)	Not reported
[11]	90nm CMOS	0.5-7▷	18	4.5-5.5	-3	16* 4-28**	1.2	CG-CS (noise-canceling)	0.01†
[33]	90nm CMOS	0.8-6	3-36 (variable-gain)	5-5.5	-3.5	29*	2.5	CG-CS (noise-canceling)	3.8†
This Work	**0.13μm CMOS**	**1-5.2◁**	**22.4-24.3^**	**6.5-8.3^**	**≥-1.5**	**13* 35****	**1.2* 1.5****	**CG**	**1.1†**

*RX portion (including baseband)
**LO portion
***Baseband only portion
†Active area size
‡Single side-band NF
◇Inferred from Triple-Beat (TB) performance, where $TB = 2(T_X - IIP_3)$
◇◇Inferred from P_{1dB} performance, where IIP_3=P_{1dB}+9.6dB
▷1-dB bandwidth
◁2-dB bandwidth
^ 1.4GHz < f_{rf} < 5.2GHz

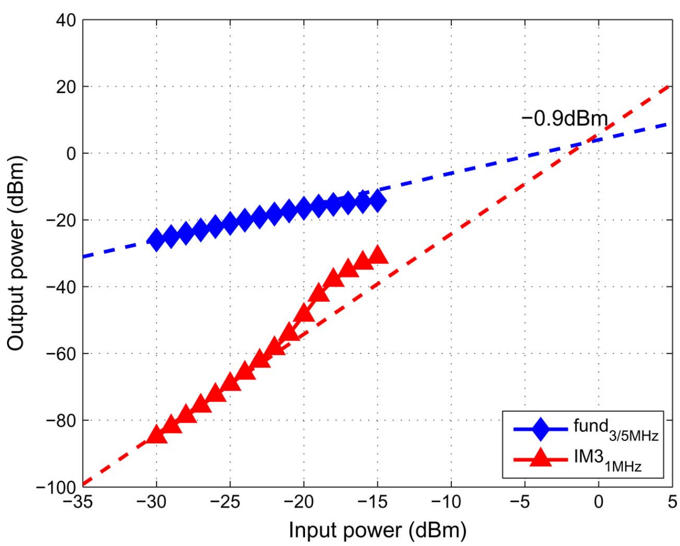


Fig. 21. Measured IIP$_3$ at $f_{\rm LO}$ = 2.7 GHz.

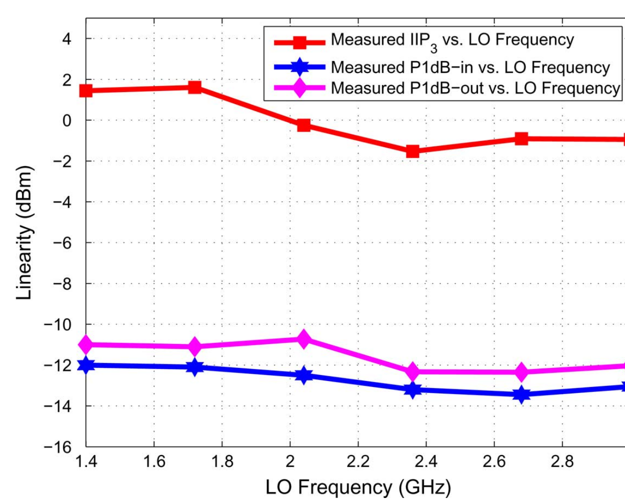


Fig. 22. Measured linearity with different $f_{\rm LO}$.

out-band linearity performance becomes significant when subsequent stages (e.g., baseband filter and a programmable-gain amplifier) are interfaced. The measured performance of the proposed receiver is summarized in Table I, and compared with recently published works with the similar architectures. The proposed broadband receiver shows outstanding performance and proves the effectiveness of the presented architecture and design methodology with low power consumption.

V. CONCLUSION

This paper has presented a broadband CMOS direct-conversion quadrature receiver architecture fabricated in a 0.13-μm CMOS technology. The prototype receiver demonstrates that a single LNTA driving a current-mode passive mixer terminated by a low-input-impedance TIA can achieve a high and flat gain with low NF over a wide bandwidth. Large voltage swing due to I–V conversion is deferred until the output of TIA and for that reason the receiver shows improved OOB tolerance and in-band linearity performance. In addition, by maintaining low-impedance nodes along the RF signal path, the receiver operates over a wide range of frequencies, and the bandwidth limitation comes mainly from insufficient LO swing.

The design tradeoffs between input impedance, transconductance, and the load impedance of broadband LNTA is overcome by employing the proposed CG LNTA with dual feedback. With the proposed topology, transconductance (G_m) is no longer limited by the source impedance (R_S) and, therefore,

Authorized licensed use limited to: ANNA UNIVERSITY. Downloaded on May 22,2025 at 05:38:35 UTC from IEEE Xplore. Restrictions apply.

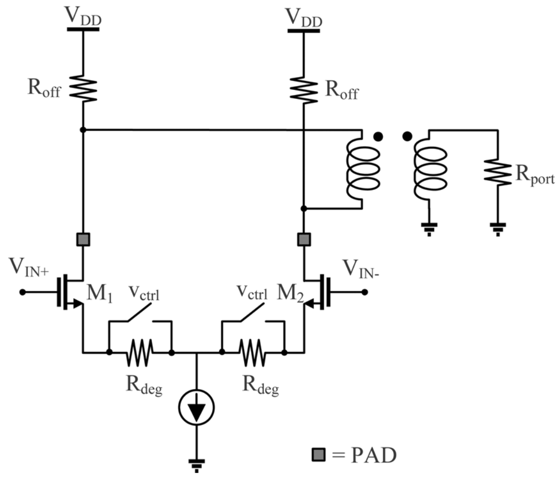


Fig. 23. Schematic of output buffer and matching network.

improved LNTA current gain enables large conversion gain and low NF over a wideband. Furthermore, the proposed LO generation scheme requires neither 50% duty-cycle $2f_{\rm LO}$ nor a 25% duty-cycle $f_{\rm LO}$, because we overlap two 50% I-Q LO signals (generated by divide-by-two) with quarter period delay. A baseband feed-forward compensation scheme is employed to achieve wide-band low input impedance and improves single-pole TIA's filtering. Measurement results demonstrate 24.3 dB maximum conversion gain, 6.5 dB minimum NF, and over -1.5 dBm $\rm IIP_3$, while dissipating 13 mW from the RF and baseband signal path from a 1.2-V supply. Compared with recently published broadband receivers utilizing a single LNTA, the proposed architecture achieves superior noise and linearity performance with low power consumption using a main stream 0.13-μm CMOS technology.

APPENDIX A
DE-EMBEDDING CONVERSION GAIN $(A_{V,RX})$ FOR THE PROPOSED RECEIVER

Fig. 23 shows the output buffer and off-chip matching network of the proposed receiver. A differential amplifier with open-drain output is loaded with an off-chip resistor $(R_{\rm off})$ set at 50 Ω. Voltage-controlled MOSFET transistors switch on or off the source-degeneration resistor $(R_{\rm deg})$. For conversion gain and NF measurement, $R_{\rm deg}$ is shorted to minimize the noise contribution of the output buffer and matching network. The stand-alone output buffer without the matching network shows -1 dB of voltage gain with $R_{\rm deg}$ shorted. For linearity measurement, $R_{\rm deg}$ yields an additional 10 dB of attenuation and minimizes the output buffer's effect on the linearity measurement. The loaded voltage gain of the output buffer and matching network is expressed as

$$A_{V,\rm loaded} = g_m \cdot \left(R_{\rm off} \parallel \frac{R_{\rm port}}{2} \right). \tag{A1}$$

Both the $R_{\rm off}$ and $R_{\rm port}$ are 50 Ω in the measurement environment, and the loaded voltage gain is 9.5 dB lower than the unloaded voltage gain. The conversion gain reported here reflects 9.5 dB of loss due to the off-chip matching network.

ACKNOWLEDGMENT

The authors would like to thank TSMC support for chip fabrication. They would also like to thank R. Kulkarni, M. Onabajo, C. J. Park, H. J. Jeon, J. Wardlaw, and E. Pankratz for their technical discussions.

REFERENCES

[1] H. Darabi and A. Abidi, "Noise in RF-CMOS mixers: A simple physical model," *IEEE J. Solid-State Circuits*, vol. 35, no. 1, pp. 15–25, Jan. 2000.

[2] N. Kim, V. Aparin, and L. E. Larson, "A resistively degenerated wideband passive mixer with low noise figure and high $\rm IIP_2$," *IEEE Trans. Microw. Theory Tech.*, vol. 58, no. 4, pp. 820–830, Apr. 2010.

[3] W. Redman-White, D. M. Leenaerts, and W. Leenaerts, "1/f noise in passive CMOS mixers for low and zero IF integrated receivers," in *Proc. Eur. Solid-State Circuits Conf.*, Sep. 2001, pp. 41–44.

[4] N. Kim, L. E. Larson, and V. Aparin, "A highly linear SAW-less CMOS receiver using a mixer with embedded Tx filtering for CDMA," *IEEE J. Solid-State Circuits*, vol. 44, no. 8, pp. 2126–2137, Aug. 2009.

[5] Z. Ru, N. A. Moseley, E. A. M. Klumperink, and B. Nauta, "Digitally enhanced software-defined radio receiver robust to out-of-band interference," *IEEE J. Solid-State Circuits*, vol. 44, no. 12, pp. 3359–3375, Dec. 2009.

[6] M. Valla, G. Montagna, R. Castello, R. Tonietto, and I. Bietti, "A 72-mW CMOS 802.11a direct conversion front-end with 3.5-dB NF and 200-kHz 1/f noise corner," *IEEE J. Solid-State Circuits*, vol. 40, no. 4, pp. 970–977, Apr. 2005.

[7] B. R. Carlton, J. S. Duster, S. S. Taylor, and J. C. Zhan, "A 2.2 dB NF, 4.9–6 GHz direct conversion multi-standard RF receiver front-end in 90 nm CMOS," in *Proc. IEEE Radio Frequency Integr. Circuits Symp.*, Jun. 2008, pp. 617–620.

[8] Y. Feng, G. Takemura, S. Kawaguchi, and P. Kinget, "Design of a high performance 2-GHz direct-conversion front-end with a single-ended RF input in 0.13 μm CMOS," *IEEE J. Solid-State Circuits*, vol. 44, no. 5, pp. 1380–1390, May 2009.

[9] O. Ikeuchi, N. Saito, and B. Nauta, "Quadrature sampling mixer topology for SAW-Less GPS receivers in 0.18 μm CMOS," in *Proc. Symp. VLSI Circuits*, Jun. 2010, pp. 177–178.

[10] M. Brandolini, P. Rossi, D. Manstretta, and F. Svelto, "Toward multistandard mobile terminals—Fully integrated receivers requirements and architectures," *IEEE Trans. Microw. Theory Tech.*, vol. 53, no. 3, pp. 1026–1038, Mar. 2005.

[11] S. C. Blaakmeer, E. A. M. Klumperink, D. M. W. Leenaerts, and B. Nauta, "THE BLIXER a wideband Balun-LNA-I/Q-Mixer topology," *IEEE J. Solid-State Circuits*, vol. 43, no. 12, pp. 2706–2715, Dec. 2008.

[12] H. Khatri, P. S. Gudem, and L. E. Larson, "Distortion in current commutating passive CMOS downconversion mixers," *IEEE Trans. Microw. Theory Tech.*, vol. 57, no. 11, pp. 2671–2681, Nov. 2009.

[13] H. T. Friis, "Noise figure of radio receivers," in *Proc. IRE*, Jul. 1944, pp. 419–422.

[14] J.-H. Zhan and S. S. Taylor, "A 5 GHz resistive-feedback CMOS LNA for low-cost multi-standard applications," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2006, pp. 721–730.

[15] J. Kim, S. Hoyos, and J. Silva-Martinez, "Wideband common-gate CMOS LNA employing dual negative feedback with simultaneous noise, gain, and bandwidth optimization," *IEEE Trans. Microw. Theory Tech.*, vol. 58, no. 9, pp. 2340–2351, Sep. 2010.

[16] D. J. Allstot, X. Li, and S. Shekhar, "Design considerations for CMOS low-noise amplifiers," in *Proc. IEEE Radio Frequency Integr. Circuits Symp.*, Jun. 2004, pp. 97–100.

[17] W. Zhuo, X. Li, S. Shekhar, S. H. K. Embabi, J. P. de Gyvez, D. J. Allstot, and E. Sánchez-Sinencio, "A capacitor cross-coupled common-gate low-noise amplifier," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 52, no. 12, pp. 875–879, Dec. 2005.

[18] A. Liscidini, M. Brandolini, D. Sanzogni, and R. Castello, "A 0.13 μm CMOS front-end for DCS1800/UMTS/802.11b-g with multi-band positive feedback LNA," *IEEE J. Solid-State Circuits*, vol. 41, no. 4, pp. 981–988, Apr. 2006.

Authorized licensed use limited to: ANNA UNIVERSITY. Downloaded on May 22,2025 at 05:38:35 UTC from IEEE Xplore. Restrictions apply.

[19] S. Woo, W. Kim, C. Lee, K. Lim, and J. Laskar, "A 3.6 mW differential common-gate CMOS LNA with positive-negative feedback," in *IEEE Int. Solid-State Circuits Conf. Dig. Tech. Papers*, Feb. 2009, pp. 218–219.

[20] N. Poobuapheun, W.-H. Chen, Z. Boos, and A. M. Kiknezad, "A 1.5-V 0.7–2.5-GHz CMOS quadrature demodulator for multiband direct-conversion receivers," *IEEE J. Solid-State Circuits*, vol. 42, no. 8, pp. 1669–1677, Aug. 2007.

[21] Y. Cheng, C.-H. Chen, M. Matloubian, and M. J. Deen, "High-frequency small signal ac and noise modeling of MOSFETs for RF IC design," *IEEE Trans. Electron Devices*, vol. 49, no. 3, pp. 400–408, Mar. 2002.

[22] G. Gonzalez, *Microwave Transistor Amplifiers: Analysis and Design*, 2nd ed. Upper Saddle River, NJ, USA: Prentice-Hall, 1997.

[23] D. Manstretta, M. Brandolini, and F. Svelto, "Second-order intermodulation mechanisms in CMOS downconverters," *IEEE J. Solid-State Circuits*, vol. 38, no. 3, pp. 394–406, Mar. 2003.

[24] R. S. Pullela, T. Sowlati, and D. Rozenbilt, "Low flicker-noise quadrature mixer topology," in *IEEE Int. Solid-State Circuits Conf. Dig. Tech. Papers*, Feb. 2006, pp. 1870–1879.

[25] A. Shahani, D. K. Shaeffer, and T. H. Lee, "A 12-mW wide dynamic range CMOS front-end for a portable GPS receiver," *IEEE J. Solid-State Circuits*, vol. 32, no. 12, pp. 2061–2070, Dec. 1997.

[26] J.-H. Zhan, B. R. Carlton, and S. S. Taylor, "A broadband low-cost direct-conversion receiver front-end in 90 nm CMOS," *IEEE J. Solid-State Circuits*, vol. 43, no. 5, pp. 1132–1137, May 2008.

[27] P. R. Gray, P. J. Hurst, S. H. Lewis, and R. G. Meyer, *Analysis and Design of Analog Integrated Circuits*, 4th ed. New York, NY, USA: Wiley, 2001.

[28] B. Razavi, *Design of Analog CMOS Integrated Circuits*. New York, NY, USA: McGraw-Hill, 2001.

[29] J. Crols and M. S. J. Steyaert, "A 1.5-GHz highly linear CMOS downconversion mixer," *IEEE J. Solid-State Circuits*, vol. 30, no. 7, pp. 736–742, Jul. 1995.

[30] B. K. Thandri and J. Silva-Martinez, "A robust feedforward compensation scheme for multistage operational transconductance amplifiers with no miller capacitors," *IEEE J. Solid-State Circuits*, vol. 38, no. 2, pp. 237–243, Feb. 2003.

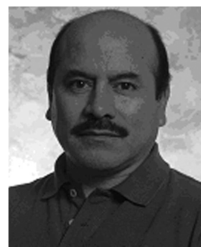
[31] B. Razavi, *Design of Integrated Circuits for Optical Communications*. New York, NY, USA: McGraw-Hill, 2002.

[32] A. Amer, E. Hegazi, and H. F. Ragaie, "A 90-nm wideband merged CMOS LNA and mixer exploiting noise cancellation," *IEEE J. Solid-State Circuits*, vol. 42, no. 2, pp. 323–328, Feb. 2007.

[33] R. Bagheri, A. Mizraei, S. Chehrazi, M. E. Heidari, M. Lee, M. Mikhemar, W. Tang, and A. Abidi, "An 800-MHz–6 GHz software-defined wireless receiver in 90-nm CMOS," *IEEE J. Solid-State Circuits*, vol. 41, no. 12, pp. 2860–2876, Dec. 2006.

Jusung Kim (S'10–M'12) received the B.S. degree (with highest honors) from Yonsei University, Seoul, Korea, in 2006, and the Ph.D. degree from Texas A&M University, College Station, TX, USA, in 2011, both in electrical engineering.

In the summer of 2008, he was an Analog IC Design Engineer with Texas Instruments, Dallas, TX, USA, where he designed an RF front-ends for multi-standard analog and digital TV silicon tuners. Since January 2011, he has been with Qualcomm Inc., San Diego, CA, USA, where he designs RFIC products for 3G and 4G cellular systems. His research interests include transceiver system and circuit design at RF and millimeter-wave frequencies.



Jose Silva-Martinez (SM'98–F'10) received the M.Sc. degree from the Instituto Nacional de Astrofisica Optica y Electrónica, Puebla, México, in 1981, and the Ph.D. degree from the Katholieke Univesiteit Leuven, Leuven, Belgium, in 1992.

From 1981 to 1983, he was with the Electrical Engineering Department, Instituto Nacional de Astrofísica Optica y Electrónica (INAOE), Puebla, México, where he was involved with switched-capacitor circuit design. In 1983, he joined the Department of Electrical Engineering, Universidad Autónoma de Puebla, where he remained until 1993. He pioneered the graduate program on Opto-Electronics in 1992. In 1993, he rejoined the Electronics Department, INAOE, and from May 1995 to December 1998, was the Head of the Electronics Department. He was a cofounder of the Ph.D. program on Electronics in 1993. He is currently with the Department of Electrical and Computer Engineering, Texas A&M University, College Station, TX, USA, where he currently holds the position of Texas Instruments I Professor in Analog Engineering. He has authored or coauthored over 100 and 160 journal and conference papers, respectively, two books, 12 book chapters, and one U.S. patent. His current field of research is in the design and fabrication of integrated circuits for communication and biomedical applications.

Dr. Silva-Martinez has served as IEEE Circuits and Systems Society (CASS) Vice President Region-9 (1997–1998) and as an associate editor for the IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II—EXPRESS BRIEFS from 1997 to 1998 and 2002 to 2003 and an associate editor of the IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I—REGULAR PAPERS from 2004 to 2005 and 2007 to 2009 and currently serves in the board of editors of six other journals. He was the recipient of the 2005 Outstanding Professor Award by the ECE Department, Texas A&M University, coauthor of the papers that received the RF-IC 2005 and MWCAS-2011 Best Student Paper Award, coadvised in testing techniques the student who was the Winner of the 2005 Best Doctoral Thesis Award, presented by the IEEE Test Technology Technical Council of the IEEE Computer Society, and he was the recipient of the 1990 European Solid-State Circuits Conference Best Paper Award.

Authorized licensed use limited to: ANNA UNIVERSITY. Downloaded on May 22,2025 at 05:38:35 UTC from IEEE Xplore. Restrictions apply.